

PAPER_204: UQFF Dark Matter — NFW, SIDM, Rotation Curves, and Virial Theorem

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Session: 50 — grokshare7514fe.txt Full Audit

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$F_U(r,t) = \sum_{i=1}^4 U_{gi} + U_m + U_A - U_{bi}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57$

$U_{bi}(r) = \kappa \cdot [SSq] \cdot \frac{GM}{r^2}, \quad \kappa = 5.0 \times 10^{-4} \text{ day}^{-1}, \quad [SSq] = 0.57, \quad \beta_i = 0.61$

<!-- ? = 5.0e-4 day¹, [SSq] = 0.57, $\beta_i = 6.1e-1$ -->

Abstract

This paper applies the UQFF buoyancy framework to dark matter structural physics: the Navarro-Frenk-White (NFW) density profile, NFW rotation curve, self-interacting dark matter (SIDM) core formation, virial theorem mass estimation, strong gravitational lensing Einstein radius, void density evolution, and peculiar velocity. The UQFF perspective unifies these through *FUBii operators that embed DM physical expressions into Frel/E_LEP scaled buoyancy forces*, capturing the feedback between vacuum energy and CDM structure formation.

UQFF Discovery: Novel application of UQFF calibration constants ($? = 5.0 \times 10^{-4} \text{ day}^1$, $[SSq] = 0.57$) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

1. NFW Density Profile

$?_{NFW}(r) = ?_s / ((r/r_s) \cdot (1 + r/r_s)^2)$

Parameters:

$?_s$ = characteristic density (from halo mass-concentration relation)

r_s = scale radius (from NFW fit to N-body simulations)

$c = r_{vir}/r_s$ = concentration parameter ($c \sim 10$ at $z=0$, increases at higher z)

Mass enclosed:

$M(r) = 4\pi ?_s r_s^3 [\ln(1+r/r_s) - r/r_s/(1+r/r_s)]$

$F_{UBii,nfw} = -F_{rel} \times (?_{NFW}(r) / E_{LEP}) \times Q_{wave} \times (4\pi r^2 \cdot d?/dr) \times r$

$U_{m,nfw}(r) = \mu(?_{vac}) \cdot (1 - e^{-?t}) \cdot [\text{Fit to universal NFW form } ?_s/(x \cdot (1+x)^2)]$

Physical context:

NFW is universal for CDM halos (Milky Way to galaxy clusters)

UQFF: vacuum field DPM_{grav} term deepens the NFW potential well

Core-cusp tension: NFW predicts cusp $?r^{-1}$, observations often show cores

2. NFW Rotation Curve

$v(r)^2 = 4\pi G \rho_s r_s^2 [\ln(1+x) - x/(1+x)] / r$ $x = r/r_s$

Asymptotic limits:

$r \ll r_s$: $v(r) \propto r^{0.5}$ (inner rising)

$r \sim r_s$: $v(r) \sim \text{maximum}$ (peak rotation speed)

$r \gg r_s$: $v(r) \propto r^{-0.5} \cdot \ln(r)^{0.5}$ (slowly declining)

Flat rotation curves require NFW halo + baryons together:

$v_{\text{total}}^2(r) = v_{\text{bary}}^2(r) + v_{\text{NFW}}^2(r)$

$F_{\text{UBii,nfwrrot}} = F_{\text{rel}} \times (v^2(r)/G / E_{\text{LEP}}) \times Q_{\text{wave}} \times [\ln(1+x) - x/(1+x)]$

$U_{\text{m,nfwrrot}}(x) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [\text{Flat rotation for } r \gg r_s]$

Calibration: Milky Way NFW:

$\rho_s \sim 0.3 \text{ GeV/cm}^3$, $r_s \sim 20 \text{ kpc}$, $v_c \sim 220 \text{ km/s}$ at Solar circle (8 kpc)

3. SIDM Core Formation

Self-interacting DM rate:

$G = \sigma \cdot (s/m) \cdot v_{\text{rel}}$ (interaction rate)

Core formation timescale:

$t_{\text{core}} \sim (\sigma \cdot s/m)^{-1} \sim 10^{10} \cdot (\sigma/108 \text{ Mpc}^2 \text{ kpc}^{-2})^{-1} \cdot (s/m / 1 \text{ cm}^2/\text{g})^{-1} \text{ yr}$

Exponential density evolution:

$\rho_{\text{core}}(t) = \rho_{\text{init}} \cdot e^{-Gt}$ (NFW cusp converts to core when $Gt \sim 1$)

$F_{\text{UBii,sidm}} = -F_{\text{rel}} \times (G \cdot \rho_{\text{init}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times \ln(0.02N)$

$U_{\text{m,sidm}}(t) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [\text{Exponential density flattening}]$

Observational constraint: $s/m \sim 0.1\text{--}1 \text{ cm}^2/\text{g}$ (galaxy clusters, Bullet Cluster)

Planck does not exclude SIDM at this level (CDM and SIDM nearly identical on large scales)

SIDM predictions:

- Dwarf galaxies: soliton cores of radius $\sim 100 \text{ pc}$ (observed)
- Galaxy clusters: rounder, less concentrated halos
- Bullet Cluster: upper limit $s/m < 1.25 \text{ cm}^2/\text{g}$ (from offset DM centroid)

4. Virial Theorem Mass

$2K + W = 0$ (virial equilibrium for collisionless system)

$K = (3/2)M \cdot s_v^2$ (kinetic energy)

$W = -(3/5)GM^2/r_h$ (potential for uniform sphere)

Virial mass:

$M_{\text{vir}} = 2|K|/G = 3 \cdot s_v^2 \cdot r_h / G$ (for spherical system)

Cluster mass from spectroscopic s_v :

$M(< r) = 3s_v^2(r) \cdot r / G + \text{corrections for anisotropy + pressure}$

$F_{\text{UBii,vir}} = F_{\text{rel}} \times (M_{\text{vir}} / E_{\text{LEP}}) \times Q_{\text{wave}} \times (s_v^2/G) \times 3$

$U_{\text{m,vir}}(r) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [s_v^2 = GM/(3r)]$

X-ray virial:

$M_{\text{vir,X}} = 3s_{\text{X}}^2 \cdot r_h / G$ (from X-ray spectroscopy instead of optical)

$U_{\text{m,virx}}(r) = \mu(\rho_{\text{vac}}) \cdot (1 - e^{-\tau}) \cdot [\text{Matches Chandra cluster observations}]$

Numerical calibration: Coma Cluster

$s_v \sim 880 \text{ km/s}$, $r_h \sim 1 \text{ Mpc}$ $\Rightarrow M_{\text{vir}} \sim 2 \times 10^{15} M_{\odot}$

5. Strong Gravitational Lensing

Einstein radius:
 $\theta_E = \sqrt{4GM(<\theta)/c^2 \cdot D_{LS}/(D_L \cdot D_S)}$

Critical surface density:
 $S_{cr} = c^2 D_S / (4\pi G D_L \cdot D_{LS})$

Convergence: $\kappa = S/S_{cr}$

Shear: γ (traceless tidal field)

Multiple images: $\kappa = 1$ at image positions

$F_{UBii, lens} = F_{rel} \times (\theta_E / \theta_{LEP}) \times Q_{wave} \times (S_{cr} \cdot \kappa)$

$U_{m, lens}(\theta) = \mu(\theta_{vac}) \cdot (1 - e^{-\kappa \theta}) \cdot [\theta_E = \sqrt{a \cdot \theta}]$ from lensing equation

Einstein ring systems:
SDP.81 (ALMA): $z_L=0.3$, $z_S=3.04$ $\theta_E \sim 1.5''$ $M(<\theta_E) \sim 10^{11} M_\odot$

UQFF: vacuum κ correction to D_{LS} shifts θ_E by $\sim 0.1\%$

6. Void Density Evolution

Void density contrast (spherical top-hat model):
 $d_v(a) = -(3/5) \cdot (0_m \cdot a + 0_v)^{-3/2} \cdot d_{v0}$

d_{v0} = initial void underdensity

Linear theory: $d_v \propto a^{-1/2}$ in Λ -dominated epoch (voids deepen faster)

Shell-crossing: $d_v \propto -1$ marks void edge (no further underdensity growth)

$F_{UBii, voidden} = -F_{rel} \times (|d_v(a)| / \theta_{LEP}) \times Q_{wave} \times (0_m \cdot a + 0_v)^{-3/2}$

$U_{m, voidden}(a) = \mu(\theta_{vac}) \cdot (1 - e^{-\kappa \theta}) \cdot [d \propto a^{-1}]$ in matter domination

Physical context in UQFF:
Voids are dominated by vacuum energy (Λ) $\rightarrow$ UQFF's $\Lambda c^2/3$ term strongest here

$F_{UBii, voidden}$ predicts void expansion driven by vacuum buoyancy

7. Peculiar Velocity Field

Peculiar velocity from linear theory:
 $v_{pec}(r) = -(fH/3) \cdot \theta d(r') \cdot r \cdot dr'/r^2$ (spherical approximation)

Redshift space distortions (RSD):
 $v_{pec, observed} = f \cdot H \cdot r + \text{noise}$ (adds to Hubble flow)

$f \sim 0_m^{0.55}$ (growth rate approximation)

Cosmic flow from Laniakea to CMB dipole:
 $v \sim 630$ km/s toward Perseus-Pisces

$F_{UBii, pec} = F_{rel} \times (fH \cdot d(r)/3 / \theta_{LEP}) \times Q_{wave} \times (dv/dz \text{ systematic})$

$U_{m, pec}(r) = \mu(\theta_{vac}) \cdot (1 - e^{-\kappa \theta}) \cdot [\text{Spherical void: integrate Poisson}]$

8. Cluster Shock Mach Number and Merger Timescale

Shock Mach number from X-ray temperature jump:
 $M = [(\theta+1)(\theta^2/\theta_1) + (\theta-1)] / (\theta^2)$ (from Rankine-Hugoniot)

$T_2/T_1 = [2\theta M^2 - (\theta-1)] \cdot [(\theta-1)M^2 + 2] / [(\theta+1)M]^2$ (temperature jump)

Coma radio relic: $M \sim 2.5$ (from spectral index $\alpha = (M^2+1)/(M^2-1)$)

Merger crossing/dynamical timescale:
 $t_{merge} = r_{vir}/s_v = v(3r_{vir}^3/(5GM))$

$F_{UBii, mach} = F_{rel} \times (M \cdot v_s / \theta_{LEP}) \times Q_{wave} \times (T_2/T_1)$

$$F_{UBii,merg} = F_{rel} \times (t_{merge} / E_{LEP}) \times Q_{wave} \times (r_{vir}/v_c)$$

$$Um,mach(?) = \mu(?_{vac}) \cdot (1-e^{-?t}) \cdot [Matches\ Coma\ radio\ relic\ shocks\ M-2-3]$$

$$Um,merg(t) = \mu(?_{vac}) \cdot (1-e^{-?t}) \cdot [3r_{vir}/(5GM)]$$

9. Summary: UQFF DM Force Hierarchy

Scale	Dominant DM Process	F_UBii Variant	Observable
kpc (dwarf)	SIDM core formation	F_UBii,sidm	Kpc-scale DM core
kpc (Milky Way)	NFW rotation	F_UBii,nfwrot	v_c = 220 km/s
Mpc (clusters)	Virial mass	F_UBii,vir	s_v = 880 km/s (Coma)
Mpc (clusters)	Lensing	F_UBii,lens	?_E ~ 1' for clusters
100 Mpc (voids)	Void underdensity	F_UBii,voidden	d ~ -0.8 in big voids
Bulk (cosmological)	Peculiar velocity	F_UBii,pec	630 km/s Laniakea flow

10. References

grok_share_7514fe.txt lines 6096–6110 (BBCEquations items 1326–1340, 1262–1268)

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PAPER_200: Um Universal Magnetism Catalogue

Navarro, Frenk, White 1996, 1997

Spergel & Steinhardt 2000 (SIDM)

Planck 2015 lensing